ENGINEERING SPECIFICATION

Portfolio Architecture & Functionalities

Sentheepan Ellankovan ("Senthee") Personal Platform

Version: 2.0 (Exact 1:1 Match)
Reference: jmancheta.cloud
Date: October 2026
Framework: Next.js 16 + Tailwind CSS v3

1. Overview & Objective

This project is a high-fidelity, production-grade personal developer portfolio engineered as an **exact, 1:1 pixel-perfect match** to the layout, interactions, animations, and typography of jmancheta.cloud (source repo: `jm-commitz/jmfolio`), customized completely with **Sentheepan Ellankovan's** authentic background, projects, skill sets, and verified assets from `PortfolioInfos.md`.

> **Key Accomplishment:** Resolved the Tailwind styling failure where uncompiled utility classes caused fullscreen unconstrained rendering. Migrated to the exact Tailwind v3 + PostCSS engine and App Router architecture, achieving zero TypeScript and build errors across all static pages.

2. Architectural Structure

The codebase follows a clean, modular structure where each domain is decoupled into self-contained components, data schemas, and utilities:

Directory / Module	Primary Purpose	Key Components & Files
`app/`	Next.js App Router root layout, home page, and dynamic App Store routes	`layout.tsx`, `page.tsx`, `globals.css`, `projects/[slug]/page.tsx`, `icon.png`, `opengraph-image.png`
`components/hero/`	Identity profile, live GitHub avatar, verified badge, and scroll header	`Hero.tsx`, `HeroAvatar.tsx`, `ProfileBar.tsx`, `profileLinks.ts`, `AchievementToast.tsx`
`components/highlights/`	Instagram-style story bubbles with timed full-screen viewer	`Highlights.tsx`, `StoryViewer.tsx`, `highlightsData.ts`
`components/experience/`	Interactive career and internship timeline with collapsible records	`Experience.tsx`, `experienceData.ts`
`components/achievements/`	Award cards, hackathon victories, and official certifications	`Achievements.tsx`, `achievementsData.ts`
`components/availability/`	Live Colombo time clock, working status, CV grab, and WhatsApp action	`AvailabilityCard.tsx`
`components/featured/`	Embla 3D carousel spotlighting standout builds in device frames	`FeaturedProject.tsx`, `FeaturedCarousel.tsx`
`components/projects/`	Filterable project directory with dynamic tag dropdown and rows	`ProjectsList.tsx`, `ProjectRow.tsx`, `CategoryDropdown.tsx`,

Directory / Module	Primary Purpose	Key Components & Files
		`projectsData.ts`
`components/projects/appstore/`	App Store-style case study details (MacBook / iPhone device frames)	`DeviceFrame.tsx`, `ProjectGallery.tsx`, `StickyAppBar.tsx`, `ExpandableText.tsx`
`components/github/`	Live GitHub contributions heat-map grid with drag scroll	`GithubContributions.tsx`, `ContributionGraph.tsx`
`components/tools/`	Grouped technical tool-stack with dual-tone SVGs for dark/light	`Tools.tsx`, `techData.ts`
`components/tour/`	Autonomous virtual Figma cursor giving interactive speech bubble tour	`TourCursor.tsx`, `cursorParts.tsx`
`components/spotify/`	Real-time music playback and playlist rail player	`SpotifyRail.tsx`, `NowPlaying.tsx`, `visitorPlayback.ts`
`components/ui/`	Independent scroll columns, canvas dot background, and intro splash	`ScrollColumn.tsx`, `PageBackground.tsx`, `HelloSplash.tsx`, `CursorFollower.tsx`
`lib/`	Integrations for GitHub GraphQL, Spotify Web API, and Supabase presence	`github.ts`, `spotify.ts`, `supabase.ts`
`public/`	Static assets, WebP live screenshots, CV PDF, and tech stack vectors	`cv/Sentheepan_Ellankovan_CV.pdf`, `images/projects/quick-blog/`

3. The Tailwind CSS & Visual Design Engine

Root Cause of the Styling Anomaly

Earlier in the development process, Tailwind v4 packages (`@tailwindcss/postcss`) were active without theme tokens, while components relied on Tailwind v3 utility classes (e.g., `bg-card`, `text-muted-foreground`, `border-border`). This caused the PostCSS pipeline to silently bypass rules, rendering unconstrained raw HTML images.

Implementation of the Design System

The project was realigned with Tailwind CSS v3 (`tailwindcss ^3.4.19`, `postcss ^8.5.31`, `autoprefixer ^10.6.1`) and configured with:

- **CSS Variable Palette:** `--background` (#0a0a0a dark / #ffffff light), `--card` (#101010 dark / #ffffff light), `--border` (#262626 dark / #e5e5e5 light), and custom muted tones for ultra-clean readability.
- **Adaptive Typography:** DM Sans variable font loaded with fallback stacks and fine anti-aliasing.
- **Custom Border Radii:** Proportional radius multipliers (`lg`, `xl`, `2xl`, `3xl`, `4xl`) matching modern iOS squircle aesthetics.
- **Dynamic Background Canvas:** An ambient radial spotlight coupled with an HTML5 canvas dot-grid with subtle breathing opacity animations.

4. Core Interactive Functionalities

1. Independent 3-Column Desktop Grid

Desktop displays (`lg:grid-cols-[1fr_1.5fr_1fr]`) provide three independent, non-chaining scrollers via `ScrollColumn.tsx`. Each column uses `overscroll-contain` and top scroll-masking to smoothly fade content as it exits the viewport.

2. Sticky Glass ProfileBar

Once the hero scrolls past 8px, `ProfileBar.tsx` seamlessly transitions into view using Framer Motion springs and heavy `backdrop-blur-2xl`, giving immediate access to Follow, Message, and "Back to Top" triggers.

3. Instagram Story Highlights

`StoryViewer.tsx` features 4 circular highlights (About, Stack, Now, Contact). Clicking opens a full-screen mobile story viewer with touch swiping, timed progress bars, pause/resume handlers, and high-res visuals.

4. Live Colombo Time & Status

`AvailabilityCard.tsx` calculates live Asia/Colombo time (UTC+05:30) with dynamic contextual status greetings (e.g. "Morning, coffee in hand 🧉", "Heads-down building 💻") alongside 1-click CV downloads.

5. App Store-Style Project Detail View

Dynamic routing at `/projects/[slug]` mimics the Apple App Store: rounded squircle app icons, developer byline, Action Pill ("OPEN"), metadata pill strip (Category, Platform, Year, Developer, Stats), and device frames.

6. Live GitHub Activity Grid

`GithubContributions.tsx` fetches live contribution data for `@Senthee1811` (234 contributions last year), rendering a weekly calendar with interactive tooltips and smooth horizontal drag-scrolling.

7. Autonomous Tour Cursor

`TourCursor.tsx` renders a Figma-style cursor named "Senthee" that glides across key sections (Profile, CTAs, Highlights, Projects, GitHub, Spotify), typing personalized conversational bubbles.

8. Circular Reveal Theme Switcher

`FloatingThemeToggle.tsx` uses the cutting-edge View Transitions API to create an expanding circular clip-path wipe originating from the toggle button, with hydration mismatch protection.

5. Projects Showcase Data Engine

All projects in `components/projects/projectsData.ts` are populated with Sentheepan's real work, live URLs, and assets:

Project	Tags & Stack	Description & Highlights	Links & Artifacts
MatchMyStuff FEATURED	Next.js, OpenAI API, Convex, TypeScript	AI lost-and-found matching reports with vector search. Built in 24h at Cursor Buildathon (**3rd Place, Convex Track**).	Device mockups, video demo, case study stats
Quick Blog FEATURED	MERN Stack, Google Gemini AI, Tailwind	Full-stack CMS with AI writer producing structured articles from user titles. Full CRUD and MongoDB schemas.	Live Web App Real WebP screenshots (desktop + mobile)
CSE Market Alert Bot FEATURED	Python, Gemini 2.5 Flash-Lite, Asyncio	Colombo Stock Exchange intelligence bot aggregating trades and sending AI summaries & breakout alerts over Telegram.	Dashboard screenshots, alert workflow breakdown
Emotional Story Assistive System	Python, Coqui TTS, PyTorch, Docker	SLIIT final-year research: emotion-adaptive storytelling for impaired children with neural prosody TTS and stutter remediation.	Microservice architecture write-up
Studio KFM	React, Express, Node.js, MongoDB	Freelance production client platform: photography & video commercial portfolio with role-based content management.	studiokfm.com Video walkthrough demo
LMS-SAAS & DeepSeek Clone	Next.js, React, TypeScript	Learning management SaaS and streaming conversational AI interface cloning DeepSeek.	GitHub open-source repositories

6. Quality Assurance & Verification

- **Zero-Warning Production Compilation:** `npm run build` successfully validates all TypeScript types, static generation parameters, and server routes in 2.2s.
- **Asset Optimization:** Live Playwright captures from `https://quick-blog-flame.vercel.app/` converted to WebP formats, reducing bundle payload by >70%.
- **Hydration Consistency:** SSR and client-side attribute parity across View Transition toggles and locale timestamps.
- **Cross-Device Responsiveness:** Smooth fallback from 3-column desktop layout to vertical mobile stream on iOS/Android viewports.